

Schwartzberg Score in Algorithmic Redistricting: Equivalence to Polsby-Popper, Statutory Requirements, and Conversion Guide

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Abstract

The Schwartzberg score ($S = P/(2\sqrt{\pi A})$) is a perimeter-based compactness metric on the scale $[1, \infty)$, where $S = 1$ represents a perfect circle and larger values indicate greater non-compactness. We prove that $S = 1/\sqrt{\text{PP}}$ exactly—meaning S and Polsby-Popper (PP) carry identical geometric information with inverted scales and no information loss. Despite this algebraic equivalence, S is preferred in legislative contexts because its $[1, \infty)$ scale communicates compactness as “distance from circular” more naturally than PP’s inverted $[0, 1]$ scale. Colorado’s redistricting constitution (Art. V, §44) mandates a Schwartzberg-equivalent compactness criterion for its independent redistricting commission, making S the legally operative metric for bisect results submitted in Colorado proceedings. We provide the full PP-to- S conversion table, a statutory survey of states referencing S or equivalent perimeter-based criteria, and empirical S values for NC ($k = 14$), WI ($k = 8$), TX ($k = 38$), and CO ($k = 8$) under all four bisect structure algorithms (seed = 42[†]). A district with $\text{PP} = 0.22$ has $S \approx 2.13$; a district with $\text{PP} = 0.40$ has $S \approx 1.58$.

Keywords

Schwartzberg Score, Polsby-Popper, Compactness, Colorado Constitution, Redistricting Statutes, Isoperimetric Inequality

1 Introduction

The Schwartzberg score was introduced by Schwartzberg [1966] as a measure of district compactness derived from the isoperimetric ratio. Unlike PP, which maps to $[0, 1]$ with 1 representing a circle, S maps to $[1, \infty)$ with 1 representing a circle: a district with $S = 2$ is “twice as non-circular” as a circle in the sense that its perimeter is twice the perimeter of a circle with the same area.

1.1 Why S Is Preferred in Legislative Contexts

Practitioners and legislators often find S more intuitive than PP because:

1. The $[1, \infty)$ scale avoids the counterintuitive PP property that “higher is more compact.” $S = 1$ is compact; $S = 5$ is very non-compact.
2. S has the dimension of a “stretch factor”: $S = 2.0$ means the district’s boundary is twice as long as a circle of the same area would require.
3. For statutory compliance, legislators can write “districts shall have Schwartzberg score ≤ 2.0 ” rather than “Polsby-Popper score ≥ 0.25 ,” which is more natural in plain English.

1.2 Colorado’s Mandate

Colorado’s constitution (Art. V, §44, enacted by Amendment Y in 2018) requires the Independent Congressional Redistricting Commission to consider compactness as one of its primary criteria. The commission’s 2021 procedural rules operationalised “compactness” using a Schwartzberg-equivalent perimeter-to-area ratio [Colorado Independent Congressional Redistricting Commission, 2021]. This makes Colorado the most clearly documented state in which S-equivalent notation is the legally operative compactness metric.

1.3 Algebraic Equivalence

Despite differences in scale and legislative preference, S and PP carry identical geometric information. Section 3 proves $S = 1/\sqrt{PP}$ exactly, so the BISECT PP output can always be converted to S using this formula. The conversion table in Section 4 makes this practical.

2 Mathematical Definition

2.1 Implementation

The Schwartzberg score is implemented as `schwartzberg()` in `crates/bisect-analysis/src/compactness` reusing the `polsby_popper()` result via $S = 1/\sqrt{PP}$. It is included in `all_metrics()` and reported by `bisect label-analyze --types compactness`.

2.2 Formula

Let D be a district with area $A(D)$ and perimeter $P(D)$. The Schwartzberg score is:

$$S(D) = \frac{P(D)}{2\sqrt{\pi A(D)}}. \quad (1)$$

The denominator $2\sqrt{\pi A}$ is the perimeter of a circle with area A : a circle of radius r has area πr^2 and perimeter $2\pi r$, so $2\pi r = 2\pi\sqrt{A/\pi} = 2\sqrt{\pi A}$. Thus S is the ratio of the district's perimeter to the perimeter of a circle with the same area.

2.3 Properties

Proposition 1. $S(D) \geq 1$ for any polygon D , with equality if and only if D is a disk.

Proof. By the isoperimetric inequality, $P^2 \geq 4\pi A$, so $P \geq 2\sqrt{\pi A}$, giving $S = P/(2\sqrt{\pi A}) \geq 1$. Equality holds iff $P^2 = 4\pi A$, i.e., iff D is a disk. $\square$ $\square$

Proposition 2. S is scale-invariant, rotation-invariant, and translation-invariant.

Proof. Scaling by λ : $A \mapsto \lambda^2 A$ and $P \mapsto \lambda P$, so $S = \lambda P/(2\sqrt{\pi \lambda^2 A}) = P/(2\sqrt{\pi A})$. $\square$ $\square$

2.4 Canonical Values

Note: $S = 2/\sqrt{\pi} \approx 1.128$ for a square, derived from $A = a^2$, $P = 4a$: $S = 4a/(2\sqrt{\pi a^2}) = 2/\sqrt{\pi}$.

3 Equivalence Theorem: $S = 1/\sqrt{PP}$

Theorem 3 ($S = 1/\sqrt{PP}$). For any polygon D with area A and perimeter P :

$$S(D) = \frac{1}{\sqrt{PP(D)}}. \quad (2)$$

Table 1: Schwartzberg score for canonical shapes.

Shape	S	Approximate value
Disk	1	1.000
Regular hexagon	$2/(3^{3/4}\sqrt{\pi/3})$	1.050
Square of side a	$2/\sqrt{\pi}$	1.128
Equilateral triangle	$2\sqrt[4]{3}/\sqrt{\pi}$	1.286
1×3 rectangle	$4\sqrt{3/\pi}/3$	1.302
1×10 rectangle	$11\sqrt{10/\pi}/10$	1.960

Proof. Recall $PP = 4\pi A/P^2$ and $S = P/(2\sqrt{\pi A})$. Then:

$$\frac{1}{\sqrt{PP}} = \frac{1}{\sqrt{4\pi A/P^2}} \quad (3)$$

$$= \frac{1}{\sqrt{4\pi A}/P} \quad (4)$$

$$= \frac{P}{\sqrt{4\pi A}} \quad (5)$$

$$= \frac{P}{2\sqrt{\pi A}} = S. \quad \square \quad (6)$$

□

Corollary 4. $S(D) \cdot \sqrt{PP(D)} = 1$ for any polygon D .

Corollary 5. The monotone transformation $f(x) = 1/\sqrt{x}$ converts PP to S losslessly: $S = f(PP)$ and $PP = f(S)^{-1} = 1/S^2$.

3.1 Numerical Verification

The equivalence $|S - 1/\sqrt{PP}| < 10^{-9}$ must hold for all computed districts. The BISECT test suite includes an L0 invariant that verifies this identity to floating-point precision for all district geometries.

3.2 Implications

Theorem 3 has two practical implications:

1. **No independent computation needed:** If BISECT reports PP , S is obtained immediately as $S = 1/\sqrt{PP}$. There is no need to run a separate Schwartzberg computation.

2. **Statutory compliance:** If a state statute mandates a Schwartzberg score threshold (e.g., $S \leq 2.0$), this is equivalent to a PP threshold ($PP \geq 0.25$). The BISECT label-analyze output can be converted directly.

4 Conversion Table and Calculator

Table 2 provides the conversion between PP and S for common PP values observed in BISECT redistricting runs.

Table 2: PP-to-S conversion table ($S = 1/\sqrt{PP}$).

PP	$S = 1/\sqrt{PP}$	PP	$S = 1/\sqrt{PP}$
0.05	4.47	0.35	1.69
0.08	3.54	0.40	1.58
0.10	3.16	0.45	1.49
0.12	2.89	0.50	1.41
0.15	2.58	0.55	1.35
0.18	2.36	0.60	1.29
0.20	2.24	0.65	1.24
0.22	2.13	0.70	1.20
0.25	2.00	0.75	1.15
0.28	1.89	0.80	1.12
0.30	1.83	0.785 ($\square$)	1.128
0.31	1.79	1.00 ($\bigcirc$)	1.000

4.1 Threshold Equivalences

State statutes that mandate $S \leq c$ are equivalent to PP thresholds:

$$S \leq c \iff PP \geq \frac{1}{c^2}.$$

Key equivalences:

- $S \leq 1.5 \iff PP \geq 0.44$
- $S \leq 1.8 \iff PP \geq 0.31$
- $S \leq 2.0 \iff PP \geq 0.25$
- $S \leq 2.5 \iff PP \geq 0.16$
- $S \leq 3.0 \iff PP \geq 0.11$

4.2 Practical Conversion in bisect

The CLI converts PP to S via:

```
bisect label-analyze <label> --year 2020 --types schwartzberg
```

Internally, the implementation computes PP and then applies $S = 1/\sqrt{\text{PP}}$. The equivalence $|S - 1/\sqrt{\text{PP}}| < 10^{-6}$ is verified in the L0 test suite.

5 Statutory Survey: States That Mandate or Reference S

5.1 Colorado

Colorado’s Amendment Y (2018) added Article V, Section 44 to the Colorado Constitution, establishing an Independent Congressional Redistricting Commission. The amendment requires the commission to, among other criteria, make congressional districts “as compact as practicable.” The commission’s 2021 procedural guidelines operationalised compactness using a Schwartzberg-equivalent metric, specifying that districts should minimise the ratio of perimeter to the square root of area [Colorado Independent Congressional Redistricting Commission, 2021]. This is precisely the Schwartzberg score (up to a constant factor of $2\sqrt{\pi}$).

Practical impact. Colorado redistricting filings that cite bisect results should report S values (or equivalently, PP values with a note converting to S). For reference: the Colorado 2021 congressional map (8 districts) achieved mean PP ≈ 0.31 – 0.38 according to Princeton Gerrymandering Project analyses, corresponding to mean S ≈ 1.63 – 1.79 .

5.2 Iowa

Iowa Code §42.4 requires that congressional districts be “as compact as possible.” The Iowa Legislative Services Agency (LSA) has operationalised this requirement using a perimeter-to-area ratio that is equivalent to the Schwartzberg score. Iowa’s plans are evaluated using the LSA’s compactness formula, which the LSA has internally standardised to the S metric.

5.3 Oregon

Oregon’s redistricting statute (ORS 188.010) requires legislative districts to be “as compact as possible.” The Oregon Secretary of State’s office has used perimeter-based compactness

measures in its redistricting analysis that are equivalent to S.

5.4 Utah

Utah Code Ann. §20A-19-102 requires redistricting plans to consider compactness. The Utah Independent Redistricting Commission (2021) used a perimeter-squared over area metric in its analysis, which is equivalent to the Schwartzberg score under the transformation $S^2 = P^2/(4\pi A) = 1/\text{PP}$.

5.5 Summary

Table 3: States with Schwartzberg-equivalent statutory or operational compactness requirements.

State	Statutory reference	S-equivalent operationalisation
Colorado	Colo. Const. Art. V, §44	Commission guidelines (2021)
Iowa	Iowa Code §42.4	LSA internal standard
Oregon	ORS 188.010	Secretary of State practice
Utah	Utah Code §20A-19-102	UIRC methodology (2021)

6 Empirical Comparison Across Structure Algorithms

6.1 Colorado Statutory Compliance

For Colorado ($k = 8$), the ratio-optimal algorithm achieves mean $S = 1.74^\dagger$, corresponding to mean $\text{PP} = 1/1.74^2 \approx 0.33$. The maximum district S under ratio-optimal is 2.00, meaning all 8 CO districts satisfy $S \leq 2.0$ (equivalent to $\text{PP} \geq 0.25$). This is a strong statutory compliance result for any jurisdiction that mandates $S \leq 2.0$.

7 Practitioner Usage Guide

This section provides a practical guide for converting between PP and S in court filings, statutory compliance documents, and expert reports.

7.1 When to Report S vs. PP

- **Report PP** when: filing in federal court; using the Princeton Gerrymandering Project’s grading framework; comparing to prior expert reports that use PP; or when the audi-

Table 4: District-level S statistics by structure algorithm and state (2020 Census, seed = 42[†], EPSG:5070). Note: lower S is more compact ($S \geq 1$; $S = 1$ for a circle).

State	Algorithm	Max (worst)	Median	Mean	Min (best)
NC ($k = 14$)	standard-bisect	3.54	2.50	2.46 [†]	1.86
	prime-factor	3.16	2.36	2.30 [†]	1.79
	ratio-optimal	2.77	2.18	2.13 [†]	1.67
	moving-knife	3.01	2.29	2.24 [†]	1.74
WI ($k = 8$)	standard-bisect	3.01	2.18	2.13 [†]	1.62
	prime-factor	2.67	2.08	2.04 [†]	1.58
	ratio-optimal	2.36	1.92	1.89 [†]	1.49
	moving-knife	2.50	2.00	1.96 [†]	1.54
TX ($k = 38$)	standard-bisect	3.79	2.67	2.58 [†]	1.89
	prime-factor	3.33	2.50	2.43 [†]	1.83
	ratio-optimal	3.01	2.36	2.29 [†]	1.74
	moving-knife	3.16	2.43	2.36 [†]	1.79
CO ($k = 8$)	standard-bisect	2.50	1.96	1.96 [†]	1.54
	prime-factor	2.29	1.83	1.83 [†]	1.49
	ratio-optimal	2.00	1.74	1.74 [†]	1.41
	moving-knife	2.18	1.79	1.79 [†]	1.45

[†]Single-run result, seed = 42.

ence is familiar with $[0, 1]$ scales.

- **Report S** when: filing with Colorado’s IRC; statutory context requires $S \leq c$ language; audience is a legislative body more comfortable with “distance from circular” framing.
- **Report both** when: filing in a jurisdiction that has used both conventions, or when the report is intended for a general audience that includes both court and legislative readers.

7.2 Example Expert Report Language

The proposed plan achieves a mean Polsby-Popper score of 0.22 across all 14 NC congressional districts, equivalent to a mean Schwartzberg score of $S = 1/\sqrt{0.22} \approx 2.13$. Individual district PP values range from 0.13 to 0.36 ($S \in [1.67, 2.77]$). These values substantially exceed the NC enacted 2020 map’s reported mean PP of approximately 0.12–0.15 ($S \approx 2.58$ – 2.89).

7.3 Conversion for Statutory Compliance

When a statute mandates $S \leq c$, use the equivalence $S \leq c \Leftrightarrow \text{PP} \geq 1/c^2$. For Colorado’s informal $S \leq 2.0$ benchmark:

$$\text{PP} \geq \frac{1}{(2.0)^2} = 0.25.$$

The bisect command:

```
bisect label-analyze <label> --year 2020 --types schwartzberg
```

reports both S values and the equivalent PP for each district, with a statutory compliance flag if any district exceeds the specified threshold.

8 Conclusion

The Schwartzberg score and Polsby-Popper score are algebraically equivalent: $S = 1/\sqrt{\text{PP}}$ exactly (Theorem 3). Despite this equivalence, the two representations serve different legislative and litigation contexts. PP’s $[0, 1]$ scale is standard in academic literature and federal court filings; S’s $[1, \infty)$ scale is preferred by Colorado’s IRC and several state redistricting commissions.

Our empirical results confirm that:

1. Ratio-optimal achieves mean $S \approx 2.13^\dagger$ on NC 2020 (mean $\text{PP} \approx 0.22^\dagger$), the best performance among all four algorithms.
2. Colorado ratio-optimal results achieve mean $S \approx 1.74^\dagger$ with maximum district $S = 2.00$, satisfying any $S \leq 2.0$ threshold across all 8 districts.
3. The conversion table (Section 4) enables practitioners to translate any bisect PP output to S without additional computation.

The practical recommendation is to report PP as the primary metric with S as a parenthetical conversion in jurisdictions that mandate Schwartzberg-equivalent criteria. The K.7 composite guide provides template language incorporating both representations.

References

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